External-Internal Attention for Hyperspectral Image Super-Resolution

Zhiling Guo, Jingwei Xin, Nannan Wang, *Member, IEEE*, Jie Li, and Xinbo Gao, *Senior Member, IEEE*

***Abstract*— In recent years, hyperspectral image (HSI) super-resolution (SR) has made significant progress by leveraging convolution neural networks. Existing methods with spectral or spatial attention, which only consider the spectral similarity or pixel-pixel similarity, ignore sample-sample correlations and sparsity. Therefore, based on the fusion of HSI and multispectral image, we propose a new HSI SR model with external-internal attention (EIA). Instead of considering a single sample, external attention module is employed to exploit the incorporating correlations between different samples to get a better feature representation. In addition, an internal attention module based on nonlocal operation is designed to explore the long-range dependencies information. Particularly, oriented to high mapping precision and low computational cost inference, spherical locality sensitive hashing (LSH) is used to divide features into different hash buckets so that every query point is calculated in the hash bucket assigned to it, rather than based a weight sum of features across all positions. The sequential EIA greatly improves the generalization ability and robustness of the model by modeling at the dataset level and at the sample level. Extensive experiments are conducted on five widely used datasets in comparison with state-of-the-art models, demonstrating the advantage of the method we proposed.**

***Index Terms*— External-internal attention (EIA), hyperspectral image (HSI), spherical locality sensitive hashing (SLSH), super-resolution (SR).**

I. Introduction

THE energy received by a sensing unit in remote imaging is the double integration of electromagnetic waves in space and wavelength. If both spatial resolution and spectral resolution are improved, the signal is integrated over smaller ground and narrower wavelength, weakening the signal, reducing the signal-to-noise ratio and resulting in poor imaging quality. Therefore, spatial resolution and spectral resolution are balanced to ensure that the received signal is strong enough. Hyperspectral images (HSIs) have high spectral resolution and contain rich spectral information. HSIs typically have dozens to hundreds of bands that can be combined to provide information for tasks such as classification [1], [2], [3], tracking [4], [5], [6] and so on [7], while each band can be individually analyzed for specific applications. Compared to HSIs, multispectral images (MSIs) have high spatial resolution and less spectral information. However, due to the inherent hardware limitations of a hyperspectral imaging system, people can only get low-resolution HSIs (LR-HSIs) and High-resolution MSIs (HR-MSIs). As a result, HSI-MSI fusion based methods have gained increasing attention [8], [9], [10].

Pan-sharpening [11] belonging to the model-based image fusion approach aims to reconstruct HR-HSI by solving manually designed objective functions. As a special case of image fusion, LR-HSI and HR single channel panchromatic image, which capture the same scene, are fused to obtain the target image. Due to the low spectral resolution of panchromatic images, these methods [12], [13] are fast but tend to introduce spectral distortion when estimating the scene response of the latent HR HSIs from panchromatic images.

In order to avoid important spectral information distortion, some methods based on matrix decomposition are proposed for HSI super-resolution (SR). The target HSI can be linearly represented by a few spectral bases, which constitute the spectral basis matrix, and the represented coefficients constitute the spectral coefficient matrix. Kawakami *et al.* [14] used sparse prior to learn spectral bases from LR-HSI, and then performed sparse coding on HR-MSI to estimate the coefficient matrix. For a long time in the past, methods based on matrix factorization, such as [15], [16], have achieved state-of-the-art performance. Compared with methods based on pan-sharpening, these methods are easier to obtain abundant high-frequency information. However, these methods rely on prior knowledge, such as linear spectral mixture assumption and sparsity prior, which are applied under certain restrictions. Because of lacking end-to-end optimization, such hand-crafted priors with limited representative ability usually inevitably lead to performance degradation.

In recent years, deep learning-based methods, therefore, attract more and more attention. The pioneering work,

Manuscript received 1 June 2022; revised 14 July 2022 and 8 September 2022; accepted 11 September 2022. Date of publication 16 September 2022; date of current version 3 October 2022. This work was supported in part by the National Key Research and Development Program of China under Grant 2018AAA0103202; in part by the National Natural Science Foundation of China under Grant 62176195, Grant 61922066, Grant 61876142, Grant 62036007, and Grant 61976166; in part by the Technology Innovation Leading Program of Shaanxi under Grant 2022QFY01-15 and Grant 2021GY-030; in part by the Open Research Projects of Zhejiang Laboratory under Grant 2021KG0AB01; in part by the Fundamental Research Funds for the Central Universities; and in part by the Innovation Fund of Xidian University. *(Corresponding authors: Nannan Wang; Jingwei Xin.)*

Zhiling Guo, Jingwei Xin, and Nannan Wang are with the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University, Xi'an, Shaanxi 710071, China (e-mail: zhiling@stu.xidian.edu.cn; jwxin@xidian.edu.cn; nnwang@xidian.edu.cn).

Jie Li is with the School of Electronic Engineering, Xidian University, Xi'an 710071, China (e-mail: leejie@mail.xidian.edu.cn).

Xinbo Gao is with the Chongqing Key Laboratory of Image Cognition, Chongqing University of Posts and Telecommunications, Chongqing 400065, China (e-mail: gaoxb@cqupt.edu.cn).

Digital Object Identifier 10.1109/TGRS.2022.3207230

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:42:39 UTC from IEEE Xplore. Restrictions apply.

SRCNN [17] has demonstrated the advantage of convolution neural network (CNN) over conventional methods to model a complex mapping between LR images and HR images. These methods [18], [19], [20], [21] based on skip connections, including residual block (RB) [22], dense block (DB) [23] and residual DB [24], are widely used to make network even deeper and ensure good performance. Unfortunately, it is not only computationally redundant but also difficult to exploit underlying high-frequency information by repeatedly stacking convolution layers or RBs. From the perspective of prior information, the fusion-based approaches [25], [26], [27] can obtain more high-frequency spatial information from the HR auxiliary image. Zheng *et al.* [27] proposed a novel end-to-end unsupervised model for reconstructing HR-HSI from unregistered LR-HSI and HR-MSI. Due to a large number of HSI channels, its spectral information is relatively redundant. From the perspective of network structure, to pay attention to useful information, attention mechanism in [28], [29], [30] is naturally applied in HSI-SR. For example, Zheng *et al.* [30] proposed a network based on dual-attention mechanism, which includes spatial attention (SA) and channel attention (CA). The proposed channel-spatial-attention residual blocks (CSA ResBlocks) are able to adaptively exploit interchannel and interspatial relationships of feature maps. In fact, CA, SA, and some of their variants ignore the connections between samples and the efficiency of computation.

Because of the rapid increase of datasets, locality sensitive hashing (LSH) is widely adopted to index similar data. Namely, LSH is applied to solve the nearest neighbor search, which is the problem of retrieving the most similar item to a given query point. Hashing schemes provide low memory usage and are highly parallelizable. It assigns a hash bucket to the query point and only computes attention in the bucket, which prevents the SR model from attending to locations including noisy and less-informative. With all the above considerations in mind, we propose an end-to-end trainable and fusion-based network with external-internal attention (EIA) for HSI-SR. A single and simple CNN branch is designed to extract different scales of MSI information. Three different feature fusion blocks (FFBs) are used for the fusion process. Then, the EIA aims to focus on informative fusion features at dual level. The output of the fusion module and the EIA module is progressively used in the reconstruction stage through long skip connections. In summary, the main contributions of this work can be summarized as follows.

1) We propose a dual attention-driven SR (DAD-SR) network, which can efficiently exploit the spatial and spectral features from both HR-MSIs and LR-HSIs. To explore the underlying high-frequency information, the DAD-SR is equipped with a simple CNN branch without repeatedly stacking a large number of convolution blocks and a powerful EIA, which utilizes the fusion feature to assist the reconstructing of the latent high-resolution HSI.
2) Considering the connections between different samples and the different importance of feature information, EIA is designed for exploiting sample-sample correlation and pixel-pixel correlation. Due to leveraging nonlocal operation and spherical LSH (SLSH), long-range dependencies are captured by internal attention (IA), which reduces the computational cost and keeps high mapping precision. Without any bells and whistles, the proposed model that offers an improvement over the results of the state-of-the-art methods does not require any pretraining stages.

II. Related Work

A. Single HSI Super-Resolution

Single HSI SR plays an active role in reconstructing HR-HSI. Akgun *et al.* [31] proposed a linear deterministic model of the HSI acquisition process, and described the process as linear equations, which can not completely fit the process, becoming the limitation of the model-based method to reconstruct HR-HSI. To solve the problem of varied distribution modes for different classes, Huang *et al.* [32] used multidictionary-based sparse representation to model a novel SR mapping, which is robust to noise in both the learning and class allocation process. Considering the sparse characteristic of spectral unmixed and high pattern repeatability of spatial-spectral blocks, Li *et al.* [33] proposed a single HSI SR method by simultaneously combining the sparsity and the nonlocal self-similarity of the images, which maintains the spectral consistency and produces more image information. By introducing the concept of subpixel probability map, Xu *et al.* [34] designed a joint spectral-spatial subpixels mapping model aiming to obtain the probabilities of subpixels to different land cover classes. Knowing these subpixel probabilities, the HR image can be further produced. Recently, deep CNNs (DCNNs) have made a huge breakthrough in many tasks. Lei *et al.* [35] proposed LGCNet, which learns multi-level representation of remote sensing images including local and global details. Li *et al.* [36] presented a deep recursive residual network (GDRRN) equipped with a grouped recursive module, which depicts the complicated nonlinear mapping. To mine more spatial features of HSI, Li *et al.* [37] designed a mixed module to extract the potential features by 2-D/3-D convolution. To effectively combine 2-D and 3-D convolution in the network, Li *et al.* [38] proposed a HSI SR method by exploring the relationship between 2-D convolution and 3-D convolution. They, meanwhile, also solved the problem of structural redundancy by sharing spatial information.

B. HSI-MSI Fusion Based Methods

Although there are still some challenges, HSI SR has made a great process in the past two decades. Typically, the HSI SR fusion-based methods can be divided into four categories: bayesian, matrix factorization, tensor, and deep learning. Considering the effectiveness of sparsity, Akhtar *et al.* [39] proposed a generic bayesian sparse coding strategy. Wei *et al.* [40] carefully designed a sparse regularization term, relying on a decomposition of the scene on a set of dictionaries. Yokoya *et al.* [41] proposed a coupled nonnegative matrix factorization-based approach to reconstruct the HR-HSI with HR-MSI and LR-HSI. To exploit the manifold structure of the latent HSI through clustering in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:42:39 UTC from IEEE Xplore. Restrictions apply.

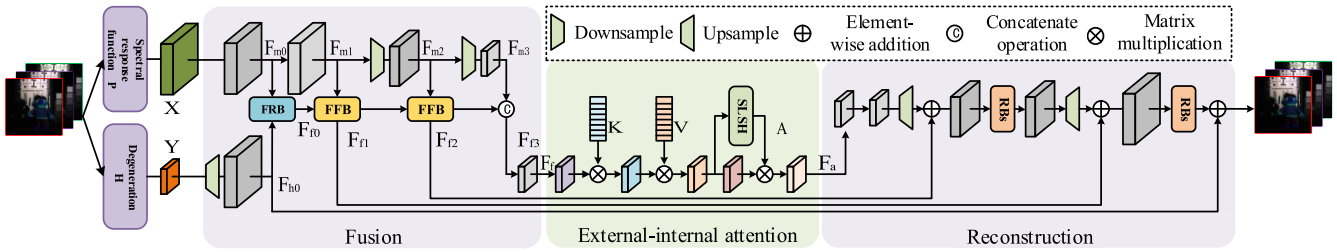


Fig. 1. Overview of the proposed DAD-SR. In order to succinctly describe the SR process, we divide it into three parts: fusion, EIA, and reconstruction. Specifically, the EIA module can integrate sample-sample correlation and pixel-pixel correlation.

spatial domain, Zhang *et al.* [42] developed an intracluster self-expressiveness model. Unlike matrix factorization-based approaches, Li *et al.* [43] designed a coupled sparse tensor factorization-based method, which considers the HSI as a 3-D tensor with three modes and factorizes it as a sparse core tensor multiplication by dictionaries of three modes. To exploit the correlation in the spectral domain, some approaches [44], [45] are proposed by making use of nonlocal similarity. To avoid intractable hand-crafted priors modeling in conventional methods, deep learning-based methods are rapidly gaining popularity. Taking consideration of the large resolution difference in the spatial domain of the observed red, green, blue (RGB) and LR-HSI, Han *et al.* [46] designed a multiscale DCNN via gradually reducing the feature size of the RGB image and increasing the feature size of LR-HSI. Zhang *et al.* [47] presented an effective CNN, which learns the deep prior from external and internal information with spatial-spectral constraint. Unsupervised learning is widely used in computer vision, such as [48], [49]. Yao *et al.* [49] proposed a coupled convolution autoencoder network (CUCaNet), which is a novel unsupervised HSI-SR model.

C. Attention Mechanism

The neural processes involved in attention have been extensively studied in neuroscience and computational neuroscience. Instead of using all available information, much of which is irrelevant for calculating neural responses, we need to select the most relevant information. Similar ideas for focusing on specific parts of input have been applied to deep learning, such as computer vision, speech recognition, machine translation, and so on. Because contextual dependency provides critical information for low-level tasks, attention mechanisms have become an important part of CNN that capture global dependencies. Many attention modules and their variants are used for HSI SR, which show great potential. Shi *et al.* [50] developed a method integrating the CA mechanism into the residual network. To explore the spectral correlation and avoid spectral distortion, Li *et al.* [29] proposed a generative network equipped with a band attention mechanism. To exploit spectral prior, Liu *et al.* [10] proposed a new spectral attention mechanism, which adaptively recalibrates features. To pay more attention to informative features, Dai *et al.* [51] proposed a second-order CA (SOCA) module to focus on discriminative representations. Take it a step further, Hu *et al.* [28] designed a mixed attention module for HSI SR by combining the first-order and second-order statistics of features.

III. Methodology

In this section, the proposed model for HSI SR is presented in detail, which aims to efficiently and effectively reconstruct HR-HSI. The overall network architecture and internal modules are shown in Fig. 1. After presenting an overview of DAD-SR in Section III-A, we introduce the details of the model in Section III-B and discuss loss function and implementation details in III-C and III-D, respectively.

A. Overview of DAD-SR

As shown in Fig. 1, in order to succinctly describe the SR process, we divide the DAD-SR model into three stages: feature fusion, EIA, and reconstruction.

Knowing two observed images, one is a LR-HSI $Y \in \mathbb{R}^{B\times n}$, consisting of B spectral bands with n pixels in the spatial domain, the other is a HR-MSI $X \in \mathbb{R}^{b\times N}$. The proposed DAD-SR based on HSI-MSI fusion aims to reconstruct a latent HR-HSI $Z \in \mathbb{R}^{B\times N}$. Generally speaking, $b < B$, $n < N$. Besides, the relation between the HR-HSI Z and the other two observations, X and Y, can be formulated as (1)

$$\begin{aligned} &X = PZ, \quad Y = ZH \\ &Y \in \mathbb{R}^{B\times n}, \quad X \in \mathbb{R}^{b\times N} \\ &Z \in \mathbb{R}^{B\times N} \\ &P \in \mathbb{R}^{b\times B}, \quad H \in \mathbb{R}^{N\times n} \end{aligned} \tag{1}$$

where $H \in \mathbb{R}^{N\times n}$ denotes the degeneration from Z to Y, and $P \in \mathbb{R}^{b\times B}$ represents the spectral response function, which is determined by the parameters of camera and can be gotten in advance. In this work, P is regarded as known [42], [43], [44], [52] and H is bicubic degeneration.

We first employ the degeneration H and spectral response function P to produce training pairs on a set of HSIs $\{Z_i\}$. Then, the SR process can be expressed with (2)

$$\begin{aligned} F_f &= F_{\mathrm{MH}}(X, Y) \\ F_a &= F_{\mathrm{EIA}}(F_f) \\ \hat{Z} &= R_{\mathrm{ec}}(F_a, F_{f2}, F_{f1}, F_{h0}) \end{aligned} \tag{2}$$

where $F_{\mathrm{MH}}(\cdot\,\cdot)$, $F_{\mathrm{EIA}}(\cdot)$ and $R_{\mathrm{ec}}(\cdot,\cdot,\cdot,\cdot)$ denotes feature fusion module, EIA module and reconstruction module, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:42:39 UTC from IEEE Xplore. Restrictions apply.

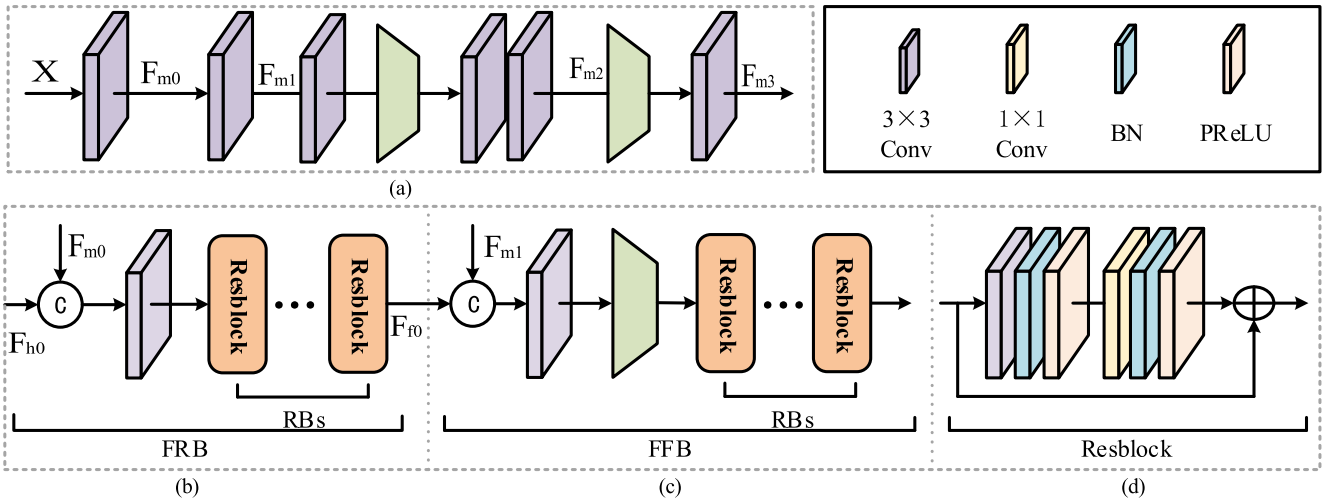


Fig. 2. Details of modules. (a) CNN branch of HR-MSI feature extraction. (b) and (c) Detail of different FFBs. Compared with FRB, FFB has an additional down-sampling operation between the convolution layer and the RBs. (d) Detail of Resblock. One RBs contains eight Resblocks.

$\hat{Z}$ is the reconstructed HR-HSI. Then, we train the proposed model with the following constraints:

$$\begin{aligned} &\min\ L_1 = 1/N_I \sum_{n=1}^{N_I} \|Z^n - \hat{Z}^n\|_1 \\ &\text{s.t. } \hat{Z}^n = \Gamma(X^n, Y^n), \quad X^n = PZ^n,\ Y^n = Z^n H \end{aligned} \tag{3}$$

where $\{Z^n, X^n, Y^n\}$ denotes the nth training pair based on degeneration H, Γ denotes the whole network model, L_1 denotes the reconstruction loss, N_I is the number of training pairs. In the following, we discuss the feature fusion module, EIA module and reconstruction module in detail.

B. Details of DAD-SR

1) Feature Fusion: The feature fusion module F_{MH} is composed of two parts, one is the feature extraction CNN branch for HR-MSI, the other is three different types of fusion blocks. The details of the CNN branch and fusion blocks can be seen in Fig. 2. Feature RB (FRB) consists of a concatenation operation, a 3×3 convolution layer and one small module RBs. One RBs contains eight Resblocks. Compared with the FRB module, FFB has an additional down-sampling operation between the convolution layer and the RBs. Except for this small difference, FRB and FFB modules include the same convolution layer and RBs. Besides, for each down-sampling operation, the length and width of the feature map become half of the original, and the up-sampling operation is the opposite. Each fusion block has two inputs. Take the first FRB, which has two inputs, F_{h0} and F_{m0}. LR-HSI Y is processed by a bilinear up-sampling and a 3×3 convolution layer to obtain feature F_{h0}. F_{m0} is derived from HR-MSI X by a simple 3×3 convolution layer. For the latter two FFBs, one input is the output of the former module, and the other is the MSI features from the CNN branch.

2) External Attention: It is considered that the attention mechanism plays an important role in various computer vision tasks. Common attention mechanisms, such as self-attention and channel-attention, usually exploit the pixel-pixel correlation or channel-channel correlation of a single sample. Particularly, for self-attention, it needs to traverse all locations. In order to solve this defect, we design an EIA module, which takes sparsity and efficiency into account.

The CA models the importance of each feature channel, and then enhances or suppresses different channels by channel weights for different tasks. For input feature X_{ca}, squeeze operation $F_{\mathrm{sq}}(X_{\mathrm{ca}})$ is carried out to compress the spatial dimension, that is, every 2-D feature becomes a real number, which is equivalent to the pooling operation with the global receptive field, and the number of feature channels remains unchanged. The excitation operation $F_{\mathrm{ex}}(X'_{\mathrm{ca}}\, W_{\mathrm{ca}})$ uses the W_{ca} parameter to generate weights for each feature channel. W_{ca} is used to explicitly model the correlation between feature channels

$$\begin{aligned} &X_{\mathrm{ca}} \Longrightarrow F_{\mathrm{sq}}(X_{\mathrm{ca}}) \Longrightarrow F_{\mathrm{ex}}\left(X'_{\mathrm{ca}}\, W_{\mathrm{ca}}\right) \\ &A_{\mathrm{ca}} = X_{\mathrm{ca}} F_{\mathrm{ex}}\left(X'_{\mathrm{ca}}\, W_{\mathrm{ca}}\right). \end{aligned} \tag{4}$$

For self-attention, we compute the attention on a number of queries simultaneously, packed together into a matrix Q. Similarly, K and V denote the keys and values matrix. As a result, we can compute the self-attention by (5)

$$\begin{aligned} A_{\mathrm{sa}} &= \mathrm{softmax}(QK^T)V \\ Q &= X_{\mathrm{sa}} W_Q \\ K &= X_{\mathrm{sa}} W_K \\ V &= X_{\mathrm{sa}} W_V \\ Q, K, V &\in \mathbb{R}^{N_{\mathrm{sa}} \times d_{\mathrm{sa}}}. \end{aligned} \tag{5}$$

It is worth noting that Q, K, and V in self-attention are derived from the same feature map X_{sa} through different layers or modules.

As shown in Fig. 1, external attention (EA) is a simple module only including some basic math operations and two fully connected layers to initialize memory units K and V. Unlike self-attention, K and V are two learnable parameters

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:42:39 UTC from IEEE Xplore. Restrictions apply.

independent of the input in EA, which is initialized by the fully connected layers. EA can calculate an attention map from the input feature X_{ea} using

$$\begin{aligned} A_{\text{ea}} &= \text{Norm}(X_{\text{ea}} M^T) \\ F_{\text{out}} &= A_{\text{ea}} M \end{aligned} \tag{6}$$

where M is a memory unit (e.g., K, V), and A_{ea} is the attention map generated from the whole training dataset. As shown in (7), in this work, K and V are chosen as two different memory units to compute the attention map

$$\begin{aligned} A_{\text{ea}} &= \text{Norm}(X_{\text{ea}} K^T) \\ F_{\text{out}} &= A_{\text{ea}} V. \end{aligned} \tag{7}$$

The processing flow of the EA is as follows:

Algorithm 1 EA Module

$F = Q_C(F_{f2}, F_{m3}) \quad \# \ F_f: \ B, C, H, W$
$A_{\text{ea}} = K(F) \quad \# \ K: \ nn.Linear(d_{\text{model}}, d_{\text{EA}})$
$A_{\text{ea}} = softmax(A_{\text{ea}}, dim = 1)$
$A_{\text{ea}} = L_1 norm(A_{\text{ea}}, dim = 3)$
$out = A_{\text{ea}} V + F \quad \# \ V: \ nn.Linear(d_{\text{EA}}, d_{\text{model}})$

For CA or self-attention, the module is modeled at the sample level, which exploits the similarity between pixels. In terms of EA, due to the introduction of memory units K and V, the module models at the dataset level, which exploits the similarity between samples. Based on learned dataset-level prior knowledge, the potential relationships between pixels in different samples benefit from improving model generalization capability and flexibility. Due to the large number of channels in HSIs, it is meaningful to explore the correlation between different samples and reduce computational costs.

3) Internal Attention: It is known that nonlocal operation produces a feature attention map by summarizing information from all positions. The IA prevents the model from attending to locations that are noisy and less informative and reduces the computational costs. For example, an input feature map $X^{H\times W\times C}$ is reshaped $X^{m\times C}$, where $m = HW$. For a query location i, the corresponding output y_i can be calculated as

$$y_i = \sum_i^m \frac{f(x_i, x_j)}{\sum_{\hat{j}=1}^m f(x_i, x_{\hat{j}})} \psi(x_j) \tag{8}$$

where x_i, x_j and $x_{\hat{j}}$, respectively, are pixel-wise features at location i, j, $\hat{j}$ on X. $\psi(\cdot)$ is a feature transformation module, and $f(x_i, x_j)$ calculates the similarity between x_i and x_j. Besides, $f(x_i, x_j) = e^{x_i^T x_j}$ is dot-product similarity, which is implementation-friendly in Pytorch. Because this is computationally expensive, it is necessary to introduce sparsity by constraining the range (not from all positions) to compute the attention map. As shown in (9)

$$y_i = \sum_{j\in\delta_i} \frac{f(x_i, x_j)}{\sum_{\hat{j}\in\delta_i} f(x_i, x_{\hat{j}})} \psi(x_j) \tag{9}$$

δ_i denotes the index set. In other words, for a query location, δ_i is the group of pixel locations IA need to be computed from. As a result, it is important to determine the δ_i for nonlocal sparse attention. Obviously, attention is to focus on the k features points that are most similar to the query point. Similarly, SLSH [53] is designed for the nearest neighbor problem.

For a Hash family, $\hbar = \{h : S \to U\}$, if $\text{pro}_1 > \text{pro}_2, r_1 < r_2$, (10) is true, $\hbar$ is called $(r_1, r_2, \text{pro}_1, \text{pro}_2)$-sensitive for any $u, v \in S$

$$\begin{aligned} \text{Pro}_\hbar[h(u) = h(v)] &\geq \text{pro}_1, \quad d(u, v) \leq r_1 \\ \text{Pro}_\hbar[h(u) = h(v)] &\leq \text{pro}_2, \quad d(u, v) > r_2 \end{aligned} \tag{10}$$

where Pro denotes probability, $h(\cdot)$ denotes a hash function, and $d(u, v)$ denotes the distance between u and v. In another word, there is a high probability that two similar vectors in a high-dimensional data space remain similar after they are mapped to a low-dimensional data space by hash function; and the two vectors that are not similar also have a large probability of not being similar in low dimensional space. Hash maps with this property are said to be locally sensitive.

The original LSH is designed to solve the following approximate nearest neighbor problem: Knowing a set of points $X \in \mathbb{R}^d$, $c = 1+\varepsilon, \varepsilon > 0, r > 0$, devise a data structure which for any query point $q \in \mathbb{R}^d$ does the following: if $d(x, q) \leq r$ return TRUE, there exists a point $\acute{x} \in X$, s.t. $d(\acute{x}, q) \leq cr$; if $d(x, q) > cr$, return FALSE for all $x \in X$.

Instead of partitioning the entire $\mathbb{R}^d$ space, SLSH is designed for partitioning the surface of the unit hypersphere in $\mathbb{R}^d$. Therefore, the conditions of the above problems should also meet $\|x\| = 1, \|q\| = 1$. In the same way, for a set of vertices $\{v_1, v_2, \ldots, v_{N_k}\}$, s.t. $\|v_i\|^2 = 1$ that forms a regular polyhedron in $\mathbb{R}^d$ where N_k denotes the number of vertices of the embedded regular polyhedron. In addition, coordinates of the vertices of the regular polyhedron in d-dimensional space are defined as follows:

$$\begin{aligned} v_{ij} &= 1 - \frac{d+1-\sqrt{d+1}}{d(d+1)}, \quad (i = j \ \& \ i = 1, 2, \ldots, N_k) \\ v_{ij} &= -\frac{d+1-\sqrt{d+1}}{d(d+1)}, \quad (i \neq j \ \& \ i = 1, 2, \ldots, N_k) \end{aligned} \tag{11}$$

where v_{ij} represents the jth coordinate of the ith vertex v_i. Let $\omega \in \mathbb{R}^{C\times N_k}$ be a rotation matrix. For a vector x, a hash function $h_\omega(x)$ is defined as the following:

$$h_\omega(x) = \arg\min_i \|\omega v_i - x\|^2. \tag{12}$$

A random rotated vector v_i can be obtained by Gram-Schmidt orthogonalization and unit length normalization. In experiments, one rotation matrix ω can be replaced simply by generating a random matrix of size $\mathbb{R}^{C\times N_k}$.

Considering the physical significance of the nearest neighbor point on the sphere, obviously, the smaller the angular distance between two vectors, the more similar the two vectors are. SLSH maps the input features onto the sphere and uses the vertices of the regular polyhedron embedded in the sphere as hash codes. As shown in Fig. 3, V_i represents one vertex of the regular polyhedron, X_i, X_j are the vectors that are mapped

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:42:39 UTC from IEEE Xplore. Restrictions apply.

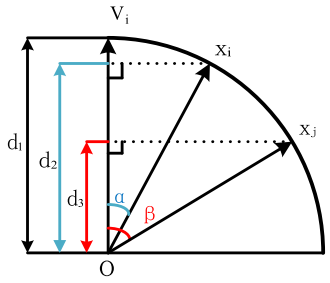


Fig. 3. O is the center of the sphere, V_i represents one vertex of a regular polyhedron, X_i, X_j are the input vectors that are mapped onto the sphere. Obviously, the smaller the angular distance between X and V_i, the more similar they are.

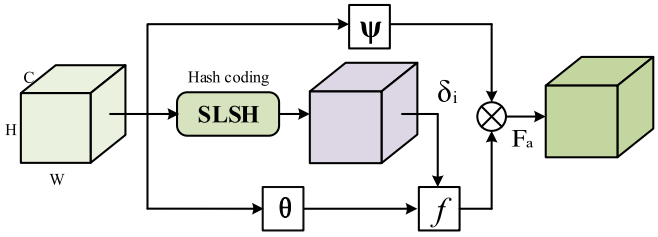


Fig. 4. Details of IA. SLSH is used to divide features into different hash buckets. θ and ψ denote different feature transformation modules. $f(\cdot,\cdot)$ measures the mutual similarity.

onto the sphere

$$\cos\alpha = \frac{\boldsymbol{OX_i}\cdot\boldsymbol{OV_i}}{|\boldsymbol{OX_i}|*|\boldsymbol{OV_i}|} = \frac{d_2}{d_1}$$
$$\cos\ \beta = \frac{\boldsymbol{OX_j}\cdot\boldsymbol{OV_i}}{|\boldsymbol{OX_j}|*|\boldsymbol{OV_i}|} = \frac{d_3}{d_1} \tag{13}$$

$\cos\ \alpha > \cos\ \beta$ means that X_i has a higher probability to choose V_i as its hash code value than X_j. As a result, (12) can be used to solve the problem in order to simplify the calculation

$$h_\omega(x) = \arg\max_i(\omega v_i x). \tag{14}$$

In this work, for IA, as shown in Fig. 4, θ and ψ denote different feature transformation modules. SLSH belongs to a special LSH function. The hash function projects a tensor onto the surface of the unit hypersphere. A hypersphere is the generalization of the sphere to a higher dimension. Then, SLSH uses a randomly rotated regular polyhedron for partitioning the surface of the unit hypersphere, and the closest polyhedron vertex v_i is chosen as its hash code. Therefore, for two similar feature vectors, they have a small angular distance on the hypersphere after SLSH, and they fall into the same attention bucket with a high probability. What is more, ω is a sampled random rotation matrix with i.i.d. Gaussian entries. N_k is also the number of hash buckets. The hash code (assigned hash bucket) is defined as $h_\omega(x) = \arg\max_i(\hat{x})$. One attention bucket of x_i can be defined by the index $\delta_i = \{j|h(x_j) = h(x_i)\}$. Thus, once the attention bucket index set δ_i for query location i is determined, the IA can be calculated using (9). The number of hash bucket $N_k = [HW/k]$ is naturally determined. IA takes linear computational complexity with respect to the input spatial size. Knowing an input feature $X \in \mathbb{R}^{n\times c}$. The whole computational cost of IA with m hash buckets is $\mathcal{O}(nm+nck+ncm)$, where $\mathcal{O}(nm)$ denotes the cost of sorting operation. The cost for attention with the number of pixels (k) in δ_i is $\mathcal{O}(nck)$. The cost of a matrix multiplication for IA with m buckets is $\mathcal{O}(ncm)$.

Similar to CA and self-attention, IA is modeled based on a single sample. In other words, same to CA and self-attention, IA mines the connections and similarities between pixels in a single sample. However, EA mines the connections and similarities between samples in a single dataset. Therefore, the EIA module can integrate sample-sample correlation and pixel-pixel correlation to obtain better feature representation ability.

4) Reconstruction: In the reconstruction stage, each RBs, which is followed by an upsampling operation, contains eight Resblocks. Besides, each Resblock consists of two Conv-batch normalization (BN)-Prelu structures, the first Conv of which is 3×3, the second is 1×1. The reconstruction process is defined as the following:

$$\begin{aligned}\hat{Z} &= R_{\text{ec}}(F_a, F_{f2}, F_{f1}, F_{h0})\\ F_a &\in \mathbb{R}^{N_r\times C_r\times H_r/4\times W_r/4}\\ F_{f2} &\in \mathbb{R}^{N_r\times C_r\times H_r/4\times W_r/4}\\ F_{f1} &\in \mathbb{R}^{N_r\times C_r\times H_r/2\times W_r/2}\\ F_{h0} &\in \mathbb{R}^{N_r\times C_r\times H_r\times W_r}\end{aligned} \tag{15}$$

where R_{ec} denotes the reconstruction stage module, F_{f2} and F_{f1} are fusion features with different sizes from the fusion stage; F_a is the output of EIA module, F_{h0} is the result of bilinear upsampling of LR-HSI. By long skip connections, F_{f2}, F_{f1} and F_{h0} continuously provide features information of different scales for reconstruction.

In summary, HR-MSI and LR-HSI are fused at multiple scales, which is beneficial to mine more detailed information. Besides, EA facilitates modeling at the dataset level, while IA with SLSH captures long rang dependencies of fused feature information and enjoys robustness and high-efficiency of sparse representation. To demonstrate the effectiveness of the EIA, we do a number of experiments to prove that DAD-SR has better modeling and generalization ability.

C. Loss Function

In recent years, L_1, L_2, perceptual, and adversarial losses are commonly used loss functions. For perceptual and adversarial losses, they can produce some details that do not exist in the original HSIs. However, in the field of remote sensing, such nonreal and nonexistent information is not what we want to get. As for L_2 loss, in order to optimize or minimize L_2 loss function, the model pays too much attention to outliers, sacrificing many normal samples. It tends to find pixel-wise averages of laten Z, which leads typically overly smooth. Besides, compared to L_1 loss, the model trained by L_2 loss constraint is relatively less robust. Due to that L_1 loss can maintain better convergence during the training phase, L_1 loss in (3) is selected to constrain the training process.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:42:39 UTC from IEEE Xplore. Restrictions apply.

TABLE I

PERFORMANCE ON CAVE DATASET AND THE DEGENERATION IS PRODUCED BY H AND 40 db NOISE

	×8							×32						
	SSPSR	NSSR	DIP	UAL	MIAE	WANG	DAD-SR	SSPSR	NSSR	DIP	UAL	MIAE	WANG	DAD-SR
RMSE ↓	8.09	8.32	3.12	1.85	2.87	1.64	**1.45**	15.40	9.05	4.98	2.66	3.95	2.72	**2.48**
PSNR ↑	31.55	32.10	38.73	43.23	41.25	45.65	**45.76**	24.50	30.00	34.83	40.43	39.73	41.82	**41.87**
SAM ↓	5.71	16.58	9.63	6.72	3.85	3.69	**3.64**	11.74	16.47	12.84	7.62	6.57	6.15	**6.11**
SSIM ↑	0.8816	0.9229	0.9646	0.9862	0.9774	0.9892	**0.9912**	0.7580	0.9194	0.9422	**0.9831**	0.9665	0.9778	0.9781

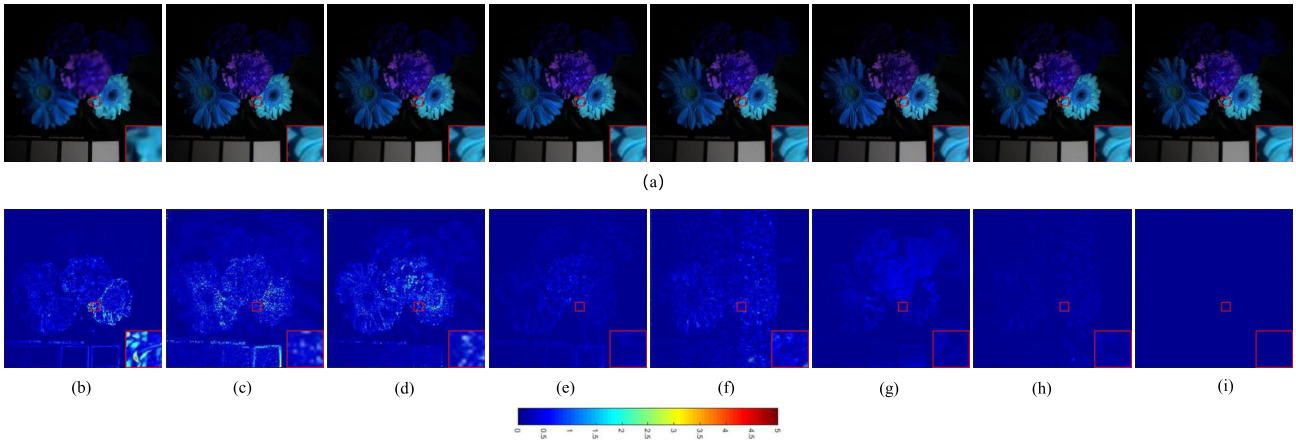


Fig. 5. Visual results and reconstruction error maps of an image in CAVE dataset with scale ×8. The first row is the reconstructed results with spectral bands 9-19-29 as RGB. The second is the corresponding error maps. (a) Reconstructed results with spectral bands 9-19-29. (b) SSPSR. (c) NSSR. (d) DIP. (e) UAL. (f) MIAE. (g) WANG. (h) DAD-SR. (i) GT.

IV. EXPERIMENTS

A. Datasets

Following [54], we use five public HSI datasets, CAVE [55], Harvard [56], ICVL [57], Chikusei dataset [58], [59] and Pavia University dataset [59] to train and test the proposed network model. CAVE dataset consists 32 HSIs, each of which is of size 512 × 512. Each image includes 31 spectral bands in a wavelength range of 400 to 700 nm at 10 nm steps. In this work, 12 HSIs are randomly selected as a training set, the remaining 20 HSIs are used to test the performance of the model. The used Harvard dataset, including 50 HSIs in a wavelength range of 420 to 720 nm, is of size 1392 × 1040, which has 31 spectral bands. 20 HSIs are used to train the model, and 30 HSIs are used for testing. The Harvard dataset includes HSIs of real-world indoor and outdoor scenes. In the ICVL dataset, we randomly select 70 HSIs each of which is of size 1392 × 1300 × 31. This dataset has a wavelength range 400–700 nm. 20 HSIs are randomly selected as a training set, the remaining 50 HSIs are used to test the performance of the model. The Chikusei dataset [58], [59] is taken by Headwall Hyperspec-visible and near-infrared camera (VNIRC) imaging sensor. It has 128 spectral bands in the spectral range from 363 to 1018 nm and 2517 × 2335 pixels in total. The Pavia University dataset [59] consists of 610 × 340 pixels with a ground sampling distance (GSD) of 1.3 m. The 12 noisy bands have been removed so a total of 103 bands. We choose the topleft area of Chikusei dataset and Pavia University dataset with a spatial size of 480 × 512 and 320 × 224 for testing.

B. Implementation Details

In experiments, we use the spectral response function of a Nikon D700 camera [42], [43], [44], [52] for CAVE, Harvard and ICVL dataset and Landsat-8 [60] for Chikusei and Pavia University dataset as P to produce an HR-MSI X. We regard the original data as the ground truth (GT) HR-HSI and simulate the LR-HSI in the same way. For the observed LR-HSI Y, the bicubic down-sampling is used as the degeneration H. The super-resolution scales are set as 8 and 32 for CAVE, Harvard and ICVL datasets. The super-resolution scale is set as eight for the Chikusei and Pavia University datasets. Given a training HSI, we extract patches of size 128 × 128 with stride 64 to generate training pairs. After LR-HSI is obtained, Gaussian white noise of different levels, 30, 35, and 40 dB, is added to it. We use the usual rotation and flip operations for data augmentation. We set the size of all Conv layers to 3 × 3 or 1 × 1. In EIA module, chunk size k is set to 32, d_{model} and d_{EA} are 512 and 128, respectively. Obviously, if $k = HW$, IA is not sparse, that is, the index set δ_i is the all positions in the feature map. For these parameter settings, we conduct corresponding ablation experiments and analysis in Section IV-C. In this work, we implement DAD-SR in Pytorch with a Titan Xp graphics processing unit (GPU). To train the proposed model, we adopt the Adam optimizer with L_1 loss. The initial learning rate is set to 1×10^{-4} and decayed to 0.9 times its original value every ten epochs until it remained unchanged at 1×10^{-6}. The batch size is set as 12 and the training procedure is terminated in 500 epochs.

C. Evaluation Criteria and Comparison Methods

Following [54], for assessing the quality of SR results on both spatial and spectral sides, we adopt four standard metrics: root-mean-square error (RMSE), peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM), and structural similarity indicator (SSIM). Generally, the larger the values

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:42:39 UTC from IEEE Xplore. Restrictions apply.

TABLE II

PERFORMANCE ON HARVARD DATASET AND THE DEGENERATION IS PRODUCED BY H AND 40 db NOISE

	×8							×32						
	SSPSR	NSSR	DIP	UAL	MIAE	WANG	DAD-SR	SSPSR	NSSR	DIP	UAL	MIAE	WANG	DAD-SR
RMSE ↓	8.34	4.18	2.57	2.08	2.15	1.67	**1.57**	14.32	4.38	3.13	2.14	2.09	1.93	**1.85**
PSNR ↑	30.35	36.28	40.43	43.00	42.62	44.39	**44.92**	28.05	35.82	38.72	41.82	42.763	43.25	**43.71**
SAM ↓	3.25	3.82	4.08	2.67	2.91	2.57	**2.50**	9.32	3.92	4.47	3.30	2.99	3.06	**2.94**
SSIM ↑	0.8372	0.9632	0.9690	0.9846	0.9836	0.9832	**0.9853**	0.7325	0.9620	0.9678	0.9787	0.9837	0.9839	**0.9842**

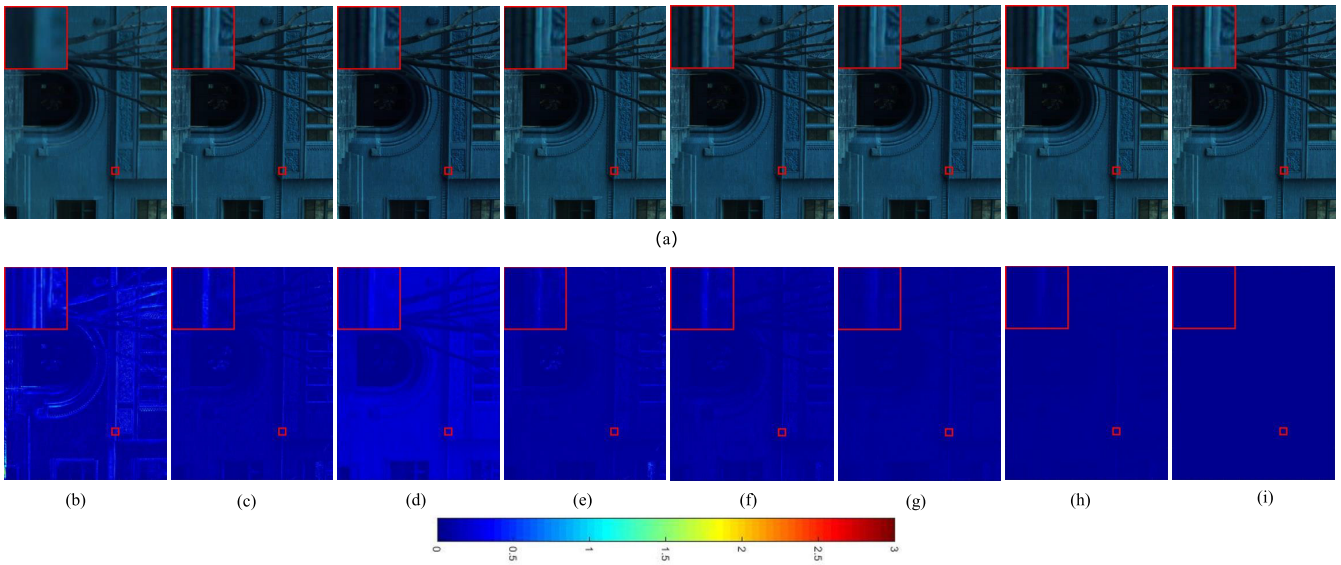


Fig. 6. Visual results and reconstruction error maps of an image in Harvard dataset with scale ×8. The first row is the reconstructed results with spectral bands 9-19-29 as RGB. The second is the corresponding error maps. (a) Reconstructed results with spectral bands 9-19-29. (b) SSPSR. (c) NSSR. (d) DIP. (e) UAL. (f) MIAE. (g) WANG. (h) DAD-SR. (i) GT.

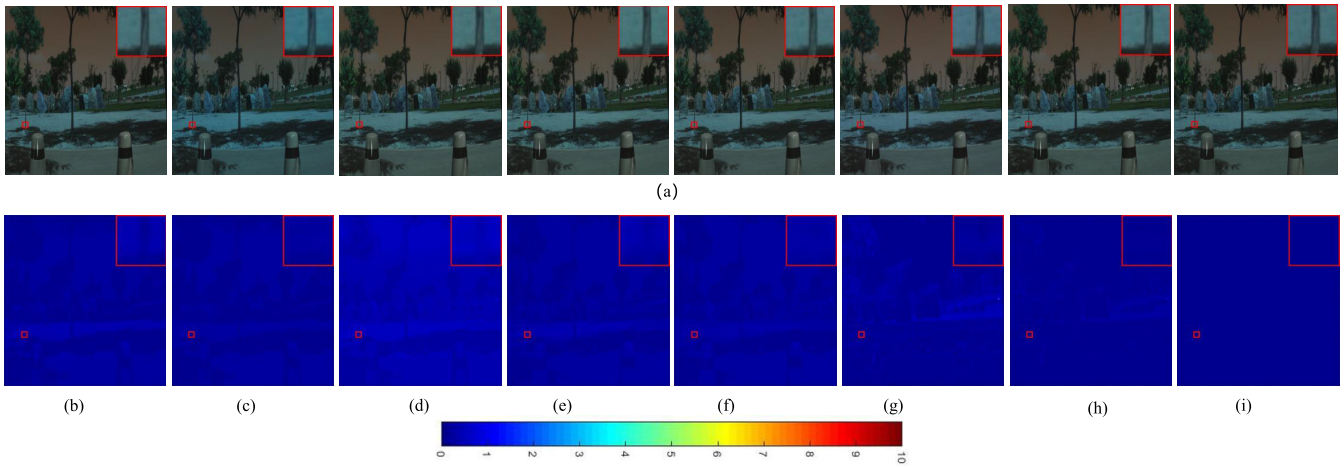


Fig. 7. Visual results and reconstruction error maps of an image in ICVL dataset with scale ×32. The first row is the reconstructed results with spectral bands 9-19-29 as RGB. The second is the corresponding error maps. (a) Reconstructed results with spectral bands 9-19-29. (b) SSPSR. (c) NSSR. (d) DIP. (e) UAL. (f) MIAE. (g) WANG. (h) DAD-SR. (i) GT.

of PSNR and SSIM, the better the spatial quality, conversely, the smaller the value of SAM, the less the spectral distortion.

In order to verify the superiority of the proposed method, we select six strategies for comparison: SSPSR [61] NSSR [44], DIP [62], UAL [54], MIAE [63], WANG [64]. Among these, SSPSR and DIP are ordinary single-image SR method; the other methods also use the corresponding spectral response function P to get HR-MSI for fusion or indirect use. In experiments, if not specified, all test images are produced by degeneration H, and 40 dB Gaussian white noise is added.

1) Result on CAVE Dataset: We compare the proposed model with the above methods, and the performance on the CAVE dataset is reported in Table I with the best values marked in bold. Obviously, the proposed DAD-SR

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:42:39 UTC from IEEE Xplore. Restrictions apply.

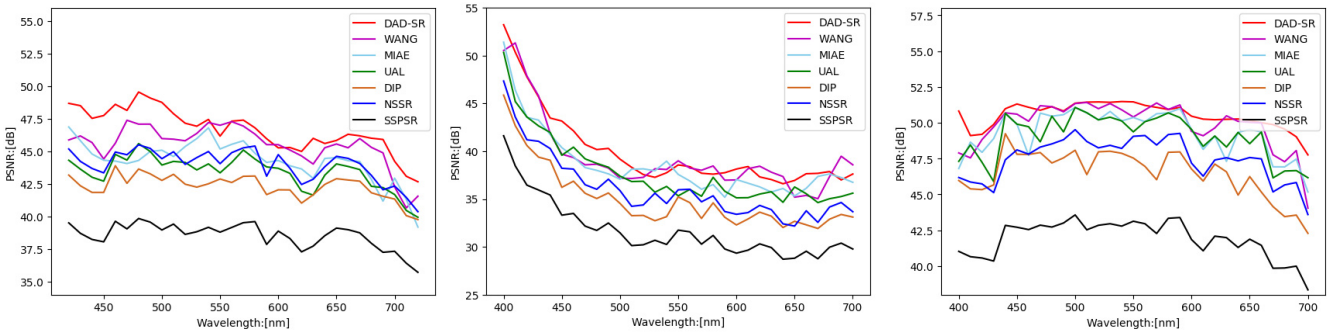


Fig. 8. Quantitative comparisons of the proposed DAD-SR with state-of-the-art methods in terms of the PSNR value of each spectral band of the reconstructed HR-HSI. (a) From CAVE dataset. (b) From Harvard dataset. (c) From ICVL dataset.

TABLE III
PERFORMANCE OF THE PROPOSED MODEL ON HARVARD DATASET WITH SCALE ×32

	PSNR↑			SAM ↓		
	img1	imgb6	imgc9	img1	imgb6	imgc9
u^2-MDN	40.97	40.25	39.59	3.95	2.94	2.68
DAD-SR	43.27	40.91	40.08	4.07	2.82	2.54

outperforms better than the other methods on all indicators. In detail, the PSNR values of the proposed model for scale ×8 and ×32 are 0.11 and 0.05 dB higher than the second best model. Besides, the reconstructed HR-HSIs and the corresponding error maps of one test image in CAVE dataset with scale ×8 are shown in Fig. 5. We select the spectral bands 9-19-29 as RGB to show the visual performance. The visual performance of these methods is relatively close, but the superiority of the proposed method can be easily seen through the corresponding error maps. The error maps corresponding to the reconstructed images of NSSR and DIP models have great differences in the zoomed region. The model performance of WANG is second only to that of our proposed method.

2) Result on Harvard: The quantitative results of the methods mentioned in Harvard dataset are shown in Table II. Compared with the fusion-based methods, SSPSR as a single HSI SR method has poor performance due to the lack of rich spatial information from MSIs. Compared to other methods, it can be observed that the method DIP has severe spatial distortion in Fig. 6. In terms of the whole spatial domain and zoomed region, the error map of the proposed method being less sharp means there is less error between it and the GT. Besides, the u^2-MDN [26] aims to reconstruct HR-HSI from unregistered or well-registered images in an unsupervised manner. We compare the proposed method with the u^2-MDN, the numerical results on the well-registered Harvard dataset with scale ×32 are reported in Table III. In terms of the SAM indicator, the two approaches are neck and neck.

3) Result on ICVL: Table IV reports the performance of the DAD-SR and other compared methods with scale ×8 and ×32. For scale ×8, the performance of DAD-SR is slightly inferior to UAL in PSNR; but for SSIM and RMSE, our method is superior. When the scale is ×32, the PSNR value of our method is 0.04 higher than the second best method. Selecting spectral bands 9-19-29 as RGB, Fig. 7 shows the reconstructed images and the corresponding error maps of a test image from the ICVL dataset. For this test image, the performance of our method obviously outperforms the other methods in error maps. Of course, other methods also can achieve quite pleasurable results.

Besides, Fig. 8 shows the PSNR value of each spectral band of images from different datasets, where the scale is ×8. Furthermore, the computational cost of different models is listed in Table V. The WANG is implemented based on UAL, and the second stage is implemented on the MATLAB platform. Therefore, the float-point operations (FLOps) of this method are more than 213.429G in qualitative analysis. Considering the overall performance, our approach achieves the best balance between performance and relatively low computing cost. In terms of the three images, the DIP method performs the worst, and DAD-SR has a slightly better performance than WANG. In summary, the superiority of the proposed method is demonstrated by numerical results, visual representation of reconstructed images, corresponding error maps, and PSNR values for each spectral band of images from three different datasets.

4) Performance on Remote Sensing Datasets: In addition, we evaluate the proposed method on two remote sensing datasets mentioned in [59]. Selecting the spectral bands 81-76-2 as RGB, Fig. 9 shows the visual exhibition of the eight times reconstructed HSI and numerical results of the Chikusei dataset. As illustrated in this figure with a meaningful region marked and zoomed in five times for easy observation, DAD-SR outperforms the others at the pixel level and PSNR. Fig. 10 shows the performance of the eight times reconstructed HSI of the Pavia University dataset. Due to the absence of MSIs and the fact that the Pavia University dataset has less training data, the performance of method SSPSR is relatively poor (25.71/6.86). As can be observed, the proposed method reconstructs more accurate results, such as the proposed method can avoid spectral distortion well, which has a small SAM value.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:42:39 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

PERFORMANCE ON ICVL DATASET AND THE DEGENERATION IS PRODUCED BY H AND 40 db NOISE

	×8							×32						
	SSPSR	NSSR	DIP	UAL	MIAE	WANG	DAD-SR	SSPSR	NSSR	DIP	UAL	MIAE	WANG	DAD-SR
RMSE ↓	9.53	3.28	4.92	1.50	1.48	1.46	**1.39**	12.95	3.34	7.57	**1.54**	1.69	1.62	1.57
PSNR ↑	30.62	38.56	34.95	**46.19**	45.87	45.92	46.09	29.08	38.45	32.83	44.56	44.76	44.98	**45.02**
SAM ↓	1.92	2.18	3.01	0.99	0.9970	0.9974	**0.9973**	2.15	2.28	4.22	1.52	1.34	1.25	**0.70**
SSIM ↑	0.8937	0.9792	0.9685	0.9963	0.9970	0.9974	**0.9973**	0.8516	0.9791	0.9586	0.9967	0.9969	0.9972	**0.9977**

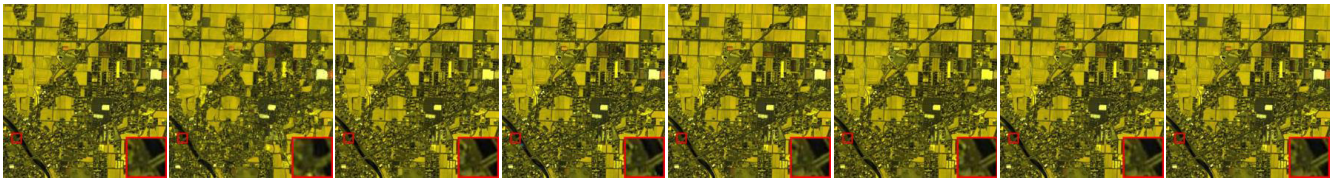


Fig. 9. Visual results of an image in Chikusei dataset with scale ×8 and spectral bands 81-76-2 as RGB.

TABLE V

NUMBER OF MODEL ANALYSIS

	SSPSR	UAL	WANG	DAD-SR
FLOPs (G)	48.974	213.429	213.429+	66.943

TABLE VI

ABLATION STUDIES OF THE PROPOSED MODEL ON HARVARD DATASET WITH SCALE ×8

		RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑
k= 32	d_{EA}= 64	1.63	44.66	2.61	0.9851
	d_{EA}=128	1.57	44.92	2.50	0.9853
	d_{EA}=192	1.65	42.58	2.64	0.9851
k= 64	d_{EA}= 64	1.84	43.87	2.80	0.9825
	d_{EA}=128	1.60	44.82	2.53	0.9853
	d_{EA}=192	1.60	44.83	2.54	0.9853
k= 96	d_{EA}= 64	1.64	44.66	2.55	0.9851
	d_{EA}=128	1.58	44.89	2.58	0.9851
	d_{EA}=192	1.61	44.76	2.55	0.9853
k=128	d_{EA}= 64	1.60	44.83	2.51	0.9852
	d_{EA}=128	2.77	42.54	3.98	0.9820
	d_{EA}=192	1.58	44.91	2.51	0.9853

TABLE VII

EFFECTIVENESS OF EACH COMPONENT ON HARVARD DATASET WITH SCALE ×8

Methods	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑
DAD-SR(Basic)	2.21	43.74	2.72	0.9839
DAD-SR w/o EA	1.61	44.73	2.56	0.9852
DAD-SR w/o IA	1.64	44.59	2.61	0.9849
DAD-SR	1.57	44.92	2.50	0.9853

D. Ablation Studies

1) Parameters of EIA Module: As shown in Algorithm 1, for memory units K and V, they are initialized by the fully connected layers. The sparsity of the IA is controlled by the size of k. In other words, if the size of k is the number of the input image pixels, there is no sparsity in IA. Therefore, we investigate the influence of different d_{EA} and k for performance. Specifically, we set $d_{\text{EA}} = 64, 128, 192$ and $k = 32, 64, 96, 128$. We do experiments on the Harvard dataset with scale ×8 to test the performance of the model under different parameters. As shown in Table VI, in terms of SSIM, there is almost no difference in the performance of these models. When $k = 32$ and $d_{\text{EA}} = 128$, the model has the best performance for PSNR, RMSE and SAM. Although the PSNR difference is only 0.01 when $k = 32$ & $d_{\text{EA}} = 128$ and $k = 128$ & $d_{\text{EA}} = 192$, the former model has fewer parameters. In this work, we select $k = 32$ and $d_{\text{EA}} = 128$. And we can see that even though k is small, the model actually gets good results, which proves IA can capture more accurate correlation information and ignore useless information.

2) Effect of Components: We remove the EIA, EA, or IA module, respectively, to obtain three variants of the proposed method which are termed "DAD-SR(Basic)," "DAD-SR w/o EA," and "DAD-SR w/o IA." Their results are given in Table VII.

To exploit the modeling capability of the proposed model at the dataset level, the EA module is introduced to model the interdependencies between samples of a dataset. To verify the effectiveness of the EA module, we compare the performance with and without EA module. We perform the experiments on the Harvard dataset, and evaluate the SR performance with a scale ×8. As shown in Table VII, with the EA mechanism, our method has achieved a slight performance gain compared to "DAD-SR w/o EA" without the EA mechanism. By adding the EA module, although the improvement of each objective indicator is relatively small, the spatial reconstruction confidence (i.e., PSNR), 44.73 versus 44.92. is more obvious than that of the improvement of spectral confidence (i.e., SAM). This proves that the EA module is conducive to exploit the representation of spatial–spectral features at the dataset level.

In addition to the above with/without EA comparisons, we also report the performance of the proposed method with/without the IA attention module. In order to verify the effectiveness of this module, we conduct experiments to verify its performance on the Harvard dataset with a scale ×8.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:42:39 UTC from IEEE Xplore. Restrictions apply.

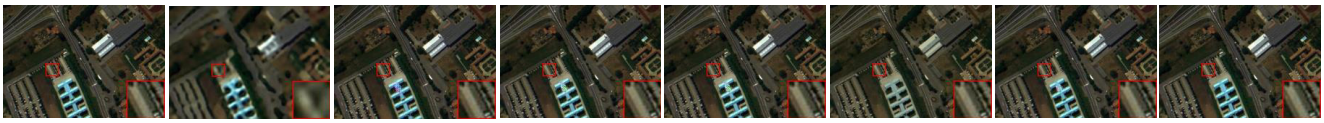


Fig. 10. Visual results of an image in Pavia University dataset with scale $\times 8$ and spectral bands 61-36-10 as RGB.

TABLE VIII
PERFORMANCE ON HARVARD DATASET AND THE DEGENERATION IS PRODUCED BY H AND 40 db NOISE

	$\times 8$						$\times 32$					
	MIAE (CAVE)	MIAE (Harvard)	MIAE (ICVL)	DAD-SR (CAVE)	DAD-SR (Harvard)	DAD-SR (ICVL)	MIAE (CAVE)	MIAE (Harvard)	MIAE (ICVL)	DAD-SR (CAVE)	DAD-SR (Harvard)	DAD-SR (ICVL)
RMSE ↓	5.16	2.15	4.44	1.67	1.57	2.21	5.87	2.09	5.13	2.44	1.85	2.22
PSNR ↑	35.90	42.62	38.59	44.31	44.92	42.44	34.66	42.76	37.32	40.87	43.71	42.02
SAM ↓	7.88	2.91	6.56	2.66	2.50	3.07	9.67	2.99	7.76	3.96	2.94	3.63
SSIM ↑	0.9500	0.9836	0.9520	0.9817	0.9853	0.9805	0.9308	0.9837	0.9533	0.9671	0.9842	0.9808

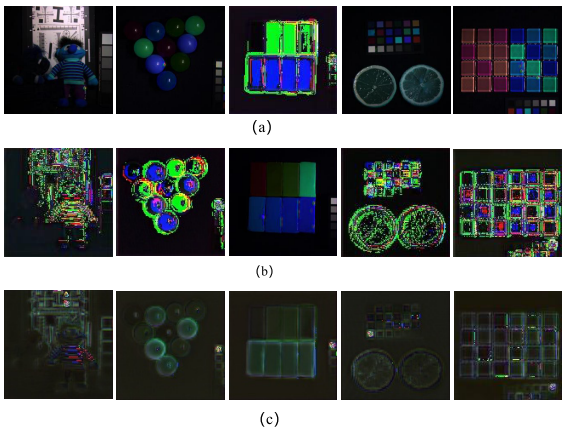


Fig. 11. Feature visualization in different stages. The first row is the reconstructed results for 480, 580, and 680 nm spectral bands. The second and third rows show the middle layer output of the DAD-SR (basic) and the DAD-SR, respectively. (a) Reconstructed result 480 nm-580 nm-680 nm. (b) Middle layer output in DAD-SR (basic). (c) Middle layer output with EIA.

As shown in Table VII, with the EA mechanism, all indicators of the model are improved, and PSNR increased by 0.33 dB.

So far, we have verified the validity of the separate EA and IA modules. Now, to verify the effectiveness of the overall EIA module, we use the basic fusion model DAD-SR (basic) to evaluate the SR performance. As shown in Fig. 11, to intuitively show the advantage of the EIA module, the intermediate maps is visualized by randomly selecting three channels. Compared with the basic DAD-SR, the proposed model can prevent the interference of worthless noise information. Besides, as shown in Table VII, removing EIA module causes obvious performance drop, e.g., PSNR drops 1.18 dB.

3) Gegeralization Ability and Robust Performance: We test the proposed method on Harvard dataset with scale $\times 8$ and $\times 32$. To demonstrate the generalization ability of the proposed DAD-SR on the Harvard dataset, we obtain three versions of DAD-SR, i.e., 'DAD-SR(CAVE)', 'DAD-SR(Harvard)' and 'DAD-SR(ICVL)', in which the model is trained on the training set of the CAVE, Harvard and ICVL datasets, respectively. We choose the Harvard testing dataset for scale $\times 8$ and $\times 32$ to indicate performance of different models. The numerical results are shown in Table VIII. It can be seen that the test performance of the models trained by different datasets does not show a huge drop on Harvard testing data for DAD-SR. In contrast, it can be seen that the generalization ability of models trained in different training sets is relatively weak for MIAE. Similarly, to verify the robustness of our model, testing images with 30, 35, and 40 dB Gaussian white noise from CAVE dataset are used to reconstruct the HR-HSI for scale $\times 8$. The detail data are shown in Table IX. For DAD-SR, it can be seen that the maximum difference of PSNR value is 1.13 dB under the influence of different noise.

E. Discussion

1) Why Does Choose SLSH: It can not be avoided the chance of far away points colliding to the near point in dimensionality reduction. However, SLSH can avoid such disadvantage by working on original input dimension, which does not require dimensionality reduction. Unlike the hypercube method [53] tends to partition the space into 2^d fragments, we use the simplex method partitioning the space into $d+1$, which supports the efficacy of SLSH. Considering efficiency and superiority, therefore, we choose this algorithm based on LSH scheme.

2) Comparison: Zheng *et al.* [30] proposed a network based on dual-attention mechanism including SA and CA. In order to take full advantage of the two kinds attention mechanisms, they integrate the SA and CA into a ResBlock. The proposed CA module exploits the interchannel relationships, which selectively suppress the less important channels and emphasize the more relevant ones. As a complement to CA module, SA module utilizes the interspatial relationships for hyperspectral pansharpening. Although we have also proposed a dual-attention mechanism for HSI SR, this is significantly different from the above work. EIA is a variant and supplement of self-attention or SA. First, EA explores the potential correlations between different samples, while self-attention or SA only considers correlation in a single sample. Then, IA aims to select the most closely related spatial regions and reduce computational cost of nonlocal attention by SLSH. In a word,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:42:39 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
PERFORMANCE UNDER DEGENERATION PRODUCED BY DIFFERENT LEVELS OF NOISE CORRUPTION AND H WITH SCALE ×8

	30 dB				35 dB				40 dB			
Methods	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑	RMSE ↓	PSNR ↑	SAM ↓	SSIM ↑
SSPSR	8.46	30.43	7.15	0.8745	8.29	31.23	6.16	0.8800	8.09	31.55	5.71	0.8816
NSSR	11.45	27.17	34.64	0.5611	9.31	29.17	25.76	0.7477	8.32	32.10	16.58	0.9229
DIP	3.64	37.16	13.95	0.9351	3.32	38.07	11.48	0.9533	3.12	38.73	9.63	0.9646
UAL	2.80	39.43	13.46	0.9566	2.23	41.59	9.36	0.9577	1.85	43.23	6.72	0.9862
MIAE	2.89	39.68	6.43	0.9772	2.88	39.80	5.20	0.9773	2.87	41.25	3.85	0.9774
WANG	2.11	44.12	6.73	0.9658	1.92	44.85	5.36	0.9602	1.64	45.65	3.69	0.9892
DAD-SR	**1.61**	**44.63**	**5.30**	**0.9853**	**1.49**	**45.41**	**4.19**	**0.9897**	**1.45**	**45.76**	**3.64**	**0.9912**

EA can be modeled at the dataset level in this work, while CA or SA only is modeled at sample level in [30], and internal attention can reduce the computational cost of SA.

V. CONCLUSION

This article presents a novel deep network including fusion module, EIA module and reconstruction module for HSI SR, which considers both long-range dependencies and sparsity. By modeling at the dataset level and sample level, the proposed model can learn more affluent features, so as to enhance the generalization ability and robustness. Particularly, SLSH efficiently selects the most informative locations to attend and reduces the computational cost. Extensive quantitative and qualitative evaluation experiments prove that the DAD-SR can obtain superior results.

REFERENCES

[1] H. Cao, X. Shang, C. Yu, M. Song, and C.-I. Chang, "Hyperspectral classification using low rank and sparsity matrices decomposition," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Oct. 2020, pp. 477–480.

[2] D. Hong, L. Gao, and N. Yokoya, "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, Aug. 2021.

[3] Y. Dong, Q. Liu, B. Du, and L. Zhang, "Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 1559–1572, 2022.

[4] J. Herweg, J. Kerekes, and M. Eismann, "Hyperspectral imaging phenomenology for the detection and tracking of pedestrians," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2012, pp. 5482–5485.

[5] B. Uzkent, A. Rangnekar, and M. J. Hoffman, "Aerial vehicle tracking by adaptive fusion of hyperspectral likelihood maps," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 39–48.

[6] H. Van Nguyen, A. Banerjee, and R. Chellappa, "Tracking via object reflectance using a hyperspectral video camera," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2010, pp. 44–51.

[7] J. Yang, Y.-Q. Zhao, J. C.-W. Chan, and S. G. Kong, "Coupled sparse denoising and unmixing with low-rank constraint for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1818–1833, Mar. 2016.

[8] Q. Xie, M. Zhou, Q. Zhao, D. Meng, W. Zuo, and Z. Xu, "Multispectral and hyperspectral image fusion by MS/HS fusion net," in *Proc. IEEE Conf. CVPR*, Jun. 2019, pp. 1585–1594.

[9] J.-F. Hu, T.-Z. Huang, and L.-J. Deng, "Hyperspectral image super-resolution via deep spatiospectral attention convolutional neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 9, 2021, doi: 10.1109/TNNLS.2021.3084682.

[10] D. Liu, J. Li, and Q. Yuan, "A spectral grouping and attention-driven residual dense network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7711–7725, Sep. 2021.

[11] L. Loncan, L. B. De Almeida, and J. M. Bioucas-Dias, "Hyperspectral pansharpening: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[12] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of MS+pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, Oct. 2007.

[13] M. Selva, B. Aiazzi, F. Butera, L. Chiarantini, and S. Baronti, "Hyper-sharpening: A first approach on SIM-GA data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3008–3024, Jun. 2015.

[14] R. Kawakami, Y. Matsushita, J. Wright, M. Ben-Ezra, Y.-W. Tai, and K. Ikeuchi, "High-resolution hyperspectral imaging via matrix factorization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2011, pp. 2329–2336.

[15] N. Akhtar, F. Shafait, and A. Mian, "Sparse spatio–spectral representation for hyperspectral image super-resolution," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2014, pp. 63–78.

[16] J. Liu, Z. Wu, L. Xiao, J. Sun, and H. Yan, "A truncated matrix decomposition for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 29, pp. 8028–8042, 2020.

[17] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2016.

[18] M. Zhang, J. Xin, J. Zhang, D. Tao, and X. Gao, "Curvature consistent network for microscope chip image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, 2022, doi: 10.1109/TNNLS.2022.3168540.

[19] M. Zhang, Q. Wu, J. Zhang, X. Gao, J. Guo, and D. Tao, "Fluid micelle network for image super-resolution reconstruction," *IEEE Trans. Cybern.*, early access, Apr. 20, 2022, doi: 10.1109/TCYB.2022.3163294.

[20] Q. Yuan, Y. Wei, X. Meng, H. Shen, and L. Zhang, "A multiscale and multidepth convolutional neural network for remote sensing imagery pan-sharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 978–989, Mar. 2018.

[21] X.-H. Han and Y.-W. Chen, "Deep residual network of spectral and spatial fusion for hyperspectral image super-resolution," in *Proc. IEEE 5th Int. Conf. Multimedia Big Data (BigMM)*, Sep. 2019, pp. 266–270.

[22] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshop*, Jun. 2017, pp. 1132–1140.

[23] T. Tong, G. Li, X. Liu, and Q. Gao, "Image super-resolution using dense skip connections," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 4809–4817.

[24] J. Xu, Y. Chae, B. Stenger, and A. Datta, "Dense ByNet: Residual dense network for image super resolution," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 71–75.

[25] W. Wei, J. Nie, L. Zhang, and Y. Zhang, "Unsupervised recurrent hyperspectral imagery super-resolution using pixel-aware refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[26] Y. Qu, H. Qi, C. Kwan, N. Yokoya, and J. Chanussot, "Unsupervised and unregistered hyperspectral image super-resolution with mutual Dirichlet-Net," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[27] K. Zheng, L. Gao, D. Hong, B. Zhang, and J. Chanussot, "NonRegSRNet: A nonrigid registration hyperspectral super-resolution network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[28] J. Hu, Y. Tang, Y. Liu, and S. Fan, "Hyperspectral image super-resolution based on multiscale mixed attention network fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[29] J. Li *et al.*, "Hyperspectral image super-resolution by band attention through adversarial learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4304–4318, Jun. 2020.

[30] Y. Zheng, J. Li, Y. Li, G. Jie, X. Wu, and J. Chanussot, "Hyperspectral pansharpening using deep prior and dual attention residual network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8059–8076, Nov. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:42:39 UTC from IEEE Xplore. Restrictions apply.

[31] T. Akgun, Y. Altunbasak, and R. M. Mersereau, "Super-resolution reconstruction of hyperspectral images," *IEEE Trans. Image Process.*, vol. 14, no. 11, pp. 1860–1875, Nov. 2005.
[32] H. Huang, J. Yu, and W. Sun, "Super-resolution mapping via multi-dictionary based sparse representation," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2014, pp. 3523–3527.
[33] J. Li, Q. Yuan, H. Shen, X. Meng, and L. Zhang, "Hyperspectral image super-resolution by spectral mixture analysis and spatial–spectral group sparsity," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 9, pp. 1250–1254, Sep. 2016.
[34] X. Xu *et al.*, "Hyperspectral image super resolution reconstruction with a joint spectral–spatial sub-pixel mapping model," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 6129–6132.
[35] S. Lei, Z. Shi, and Z. Zou, "Super-resolution for remote sensing images via local–global combined network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1243–1247, Aug. 2017.
[36] Y. Li, L. Zhang, C. Dingl, W. Wei, and Y. Zhang, "Single hyperspectral image super-resolution with grouped deep recursive residual network," in *Proc. IEEE 4th Int. Conf. Multimedia Big Data (BigMM)*, Sep. 2018, pp. 1–4.
[37] Q. Li, Q. Wang, and X. Li, "Mixed 2D/3D convolutional network for hyperspectral image super-resolution," *Remote Sens.*, vol. 12, no. 10, p. 1660, May 2020.
[38] Q. Li, Q. Wang, and X. Li, "Exploring the relationship between 2D/3D convolution for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8693–8703, Oct. 2021.
[39] N. Akhtar, F. Shafait, and A. Mian, "Bayesian sparse representation for hyperspectral image super resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 3631–3640.
[40] Q. Wei, J. Bioucas-Dias, N. Dobigeon, and J. Y. Tourneret, "Hyperspectral and multispectral image fusion based on a sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3658–3668, Jul. 2015.
[41] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.
[42] L. Zhang, W. Wei, C. Bai, Y. Gao, and Y. Zhang, "Exploiting clustering manifold structure for hyperspectral imagery super-resolution," *IEEE Trans. Image Process.*, vol. 27, no. 12, pp. 5969–5982, Dec. 2018.
[43] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.
[44] W. Dong *et al.*, "Hyperspectral image super-resolution via non-negative structured sparse representation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2337–2352, May 2016.
[45] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Nonlocal patch tensor sparse representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 3034–3047, Jun. 2019.
[46] X.-H. Han, Y. Zheng, and Y.-W. Chen, "Multi-level and multi-scale spatial and spectral fusion CNN for hyperspectral image super-resolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 4330–4339.
[47] T. Zhang, Y. Fu, L. Wang, and H. Huang, "Hyperspectral image reconstruction using deep external and internal learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Nov. 2019, pp. 8558–8567.
[48] D. Cheng, J. Zhou, N. Wang, and X. Gao, "Hybrid dynamic contrast and probability distillation for unsupervised person Re-Id," *IEEE Trans. Image Process.*, vol. 31, pp. 3334–3346, 2022.
[49] J. Yao, D. Hong, J. Chanussot, D. Meng, X. Zhu, and Z. Xu, "Cross-attention in coupled unmixing nets for unsupervised hyperspectral super-resolution," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 208–224.
[50] Z. Shi, C. W. Chen, Z. Xiong, D. Liu, Z. Zha, and F. Wu, "Deep residual attention network for spectral image super-resolution," in *Proc. Eur. Conf. Comput. Vis. Workshops*, 2018, pp. 1–16.
[51] T. Dai, J. Cai, Y.-B. Zhang, S. Xia, and L. Zhang, "Second-order attention network for single image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11057–11066.
[52] C. Lanaras, E. Baltsavias, and K. Schindler, "Hyperspectral super-resolution by coupled spectral unmixing," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2015, pp. 3586–3594.
[53] K. Terasawa and Y. Tanaka, "Spherical LSH for approximate nearest neighbor search on unit hypersphere," in *Algorithms and Data Structures* (Lecture Notes in Computer Science). Berlin, Germany: Springer, 2007, pp. 27–38.
[54] L. Zhang, J. Nie, W. Wei, Y. Zhang, S. Liao, and L. Shao, "Unsupervised adaptation learning for hyperspectral imagery super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3070–3079.
[55] F. Yasuma, T. Mitsunaga, D. Iso, and S. K. Nayar, "Generalized assorted pixel camera: Postcapture control of resolution, dynamic range, and spectrum," *IEEE Trans. Image Process.*, vol. 19, no. 9, pp. 2241–2253, Sep. 2010.
[56] A. Chakrabarti and T. Zickler, "Statistics of real-world hyperspectral images," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2011, pp. 193–200.
[57] R. Timofte *et al.*, "NTIRE 2018 challenge on single image super-resolution: Methods and results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 852–863.
[58] N. Yokoya and A. Iwasaki, "Airborne hyperspectral data over Chikusei," Space Appl. Lab., Univ. Tokyo, Japan, Tech. Rep. SAL-2016-5-27, May 2016. [Online]. Available: http://park.itc.u-tokyo.ac.jp/sal/hyperdata/TechRepSAL20160527.pdf
[59] N. Yokoya, C. Grohnfeldt, and J. Chanussot, "Hyperspectral and multispectral data fusion: A comparative review of the recent literature," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 2, pp. 29–56, Jun. 2017.
[60] A. J. Barsi, K. Lee, G. Kvaran, L. B. Markham, and A. J. Pedelty, "The spectral response of the Landsat-8 operational Land imager," *Remote Sens.*, vol. 6, no. 10, pp. 10232–10251, Oct. 2014. [Online]. Available: https://www.mdpi.com/2072-4292/6/10/10232
[61] J. Jiang, H. Sun, X. Liu, and J. Ma, "Learning spatial–spectral prior for super-resolution of hyperspectral imagery," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 1082–1096, 2020.
[62] D. Ulyanov, A. Vedaldi, and V. S. Lempitsky, "Deep image prior," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 9446–9454.
[63] J. Liu, Z. Wu, L. Xiao, and X.-J. Wu, "Model inspired autoencoder for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.
[64] X. Wang, J. Chen, and C. Richard, "Hyperspectral image super-resolution with deep priors and degradation model inversion," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 2814–2818.



Zhiling Guo received the B.Sc. degree in electronic and information engineering from the Tianjin University of Technology, Tianjin, China, in 2021. He is currently pursuing the M.S. degree with the School of Telecommunications Engineering, Xidian University, Xi'an, China.

His research interests include computer vision and machine learning.



Jingwei Xin received the B.Sc. degree in electronic information science and technology from Shaanxi Normal University, Xi'an, China, in 2016, and the Ph.D. degree in information and telecommunications engineering from Xidian University, Xi'an, in 2021.

He is currently a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. He has broad research interests in machine learning, pattern recognition, and computer vision. His research has been published in the academic venues and prestigious journals, such as European Conference on Computer Vision (ECCV), the Association for the Advancement of Artificial Intelligence (AAAI) Conference on Artificial Intelligence, and the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:42:39 UTC from IEEE Xplore. Restrictions apply.



Nannan Wang (Member, IEEE) received the B.Sc. degree in information and computation science from the Xi'an University of Posts and Telecommunications, Xi'an, China, in 2009, and the Ph.D. degree in information and telecommunications engineering from Xidian University, Xi'an, in 2015.

He is currently a Professor with the State Key Laboratory of Integrated Services Networks, Xidian University. He has published over 150 articles in refereed journals and proceedings, including the IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), International Journal of Computer Vision (IJCV), Computer Vision and Pattern Recognition (CVPR), and International Conference on Computer Vision (ICCV). His research interests include computer vision and machine learning.



Jie Li received the B.Eng. degree in electronic engineering, the M.Sc. degree in signal and information processing, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, China, in 1995, 1998, and 2004, respectively.

She is currently a Professor with the School of Electronic Engineering, Xidian University. Her research interests include image processing and machine learning. In these areas, she has published approximately 50 technical articles in refereed journals and proceedings, including the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), the IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY (TCSVT), and *Information Sciences*.



Xinbo Gao (Senior Member, IEEE) received the B.Eng., M.Sc., and Ph.D. degrees in electronic engineering, signal and information processing from Xidian University, Xi'an, China, in 1994, 1997, and 1999, respectively.

From 1997 to 1998, he was a Research Fellow with the Department of Computer Science, Shizuoka University, Shizuoka, Japan. From 2000 to 2001, he was a Post-Doctoral Research Fellow with the Department of Information Engineering, The Chinese University of Hong Kong, Hong Kong. Since 2001, he has been with the School of Electronic Engineering, Xidian University. He is also a Cheung Kong Professor of the Ministry of Education of China, a Professor of pattern recognition and intelligent system with Xidian University, and a Professor of computer science and technology with the Chongqing University of Posts and Telecommunications, Chongqing, China. He has published six books and around 300 technical articles in refereed journals and proceedings. His research interests include image processing, computer vision, multimedia analysis, machine learning, and pattern recognition.

Dr. Gao is a fellow of the Institute of Engineering and Technology and the Chinese Institute of Electronics. He served as the General Chair/Co-Chair, the Program Committee Chair/Co-Chair, or a Program Co-Chair (PC) Member for around 30 major international conferences.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:42:39 UTC from IEEE Xplore. Restrictions apply.